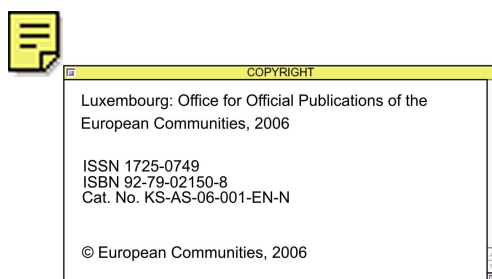


# Quality Report on International Trade Statistics



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# Quality Report on International Trade Statistics

*Eurostat – International Trade*

*2005 Edition*

## Introduction

### International trade statistics

International trade statistics, as produced by Eurostat, measure value and quantity of goods traded between Member States of the European Union (Intrastat) and value and quantity of goods traded by Member States of the EU with third countries (Extrastat). They are the official harmonised source of information about imports, exports and trade balance of the EU and of its Member States (MSs). Community legislation in the field of international trade statistics ensures that the statistics provided to Eurostat by the MSs are based on precise legal texts, directly applicable in the MSs, and on definitions and procedures which, to a large extent, have been harmonised.

The **Intrastat** system came into operation on 1 January 1993 when the Single Market was set up, causing the disappearance of the borders and of the Customs formalities for imports and exports within the EU. This system provides for direct collection of information from companies and it is based on a close link with the VAT system relating to intra-EU trade. In order to reduce the burden on traders, a series of thresholds are applied; therefore, a good quota of respondents is either exempt from submitting any declarations, or the information, which they have to provide, is significantly reduced.

Statistical information on **Extrastat** is provided by declaring parties while completing the customs formalities.

International trade statistics are an instrument of primary importance for numerous public and private sector decision makers. For example, international trade statistics:

- enable Community authorities to prepare multilateral and bilateral negotiations within the framework of the common commercial policy;
- enable Community authorities to evaluate the progress of the Single Market and the integration of EU economies;
- constitute an essential source of information for balance of payments statistics, national accounts and economic studies;
- help EU companies to do market research and define their commercial strategy.

This list, which is not exhaustive, demonstrates the diversity of the users and of their needs.

### What is quality?

Quality of official statistics in the European Statistical System is assumed to rely on elements agreed by Eurostat and MSs and is based on the following dimensions:

- Timeliness deals with the publication calendar, reference period, etc.
- Accuracy deals with the closeness between the value published and the true unknown population value, including exclusions, thresholds, non-response, adjustments, controls and corrections, confidentiality, etc.
- Accessibility deals with availability, ease of access to data, different formats and conditions of data distribution, etc.
- Clarity deals with ensuring data is adequately documented, assistance in using and interpreting the data, etc.
- Comparability deals with conceptual differences between sets of trade statistics over space and over time.

- Coherence deals with to what extent statistics originating from other sources (such as balance of payments, national accounts, etc.) are compatible with trade statistics.

### **Quality reports and indicators**

In order to monitor and analyse data quality, to provide the users of the EU international trade statistics with a tool for assessing the quality of these statistics and to propose ways to improve it, the Eurostat unit in charge of international trade statistics has started the yearly production of a Quality Report (QR). This QR is produced in accordance with the official definition of quality in the European Statistical System. Comparable national QRs on trade statistics have also been produced by most of the MSs.

The QR can be used jointly to monitor over time and improve data quality of international trade in the EU. Some indicators are based on final figures (for example, the impact of the different kind of adjustments) therefore they can only be calculated around 1 year after the end of the year they refer to. This practice explains why in some Tables of the present work data is arriving only up to year 2004. For Tables expressed in value, the annual average exchange rate is used to convert national currencies to euros.

## **1 Timeliness**

According to the Commission Regulation (EC) N. 179/2005, Member States shall transmit the monthly results of their extra-EU trade statistics to the Commission (Eurostat) no later than 40 days, in the case of aggregated results and no longer than 42 days after the end of the reference period in the case of detailed results.

For Intrastat the production of detailed data takes longer. In order to ensure optimum satisfaction of users needs and take into account of data collection and processing requirements, a distinction is made between the publication of overall results and detailed results.

According to the Commission Regulation (EC) N. 1982/2004, Member States shall forward the monthly results of intra-EU trade to the Commission (Eurostat) no later than 40 days, in the case of aggregated data, and no longer than 70 days, in the case of detailed results.

Table 1 shows each MS annual median average delay or advance, as compared with the legal deadlines, in sending intra- and extra-EU detailed data to Eurostat. This indicator is based on the dates when the first version of each monthly data is sent to Eurostat and it can therefore be calculated very quickly as compared to other indicators.

Most of the MSs, with a few exceptions, have improved the timeliness of their data during the years under analysis.

In the case of Greece, transmission of detailed data has been remarkably improved since 2003. It is also noteworthy that all new member states have managed to approach the concrete deadlines for 2005.

From the averages calculated for the European Union it follows that almost all member states conform to the European Commission regulation for 2005.

**Table 1: Transmission of detailed results - Annual median average delay (positive) or advance (negative) in working days after the legal deadline, 2003-2005**

(Legal Delay: 42 days after the reference period for extra-EU trade and 70 days for Intra-EU trade)

| MEMBER STATES  | Detailed results |      |      |                |        |      |
|----------------|------------------|------|------|----------------|--------|------|
|                | Intra-EU trade   |      |      | Extra-EU trade |        |      |
|                | 2003*            | 2004 | 2005 | 2003*          | 2004   | 2005 |
| Belgium        | -1               | -3   | -3   | -3             | -3     | -3   |
| Czech Republic |                  | -3   | -3   |                | -1     | -5   |
| Denmark        | -22              | 7    | 0    | 6              | 5      | 0    |
| Germany        | -21              | -23  | -26  | -2             | -3     | -2   |
| Estonia        |                  | 4    | -2   |                | -1     | -2   |
| Greece         | 34               | 15   | -12  | 57             | 17     | -1   |
| Spain          | -24              | -23  | -21  | -12            | -12    | -6   |
| France         | -27              | -28  | -28  | 1              | 0      | 0    |
| Ireland        | 0                | -1   | 0    | -9             | -7     | -11  |
| Italy          | 0                | 1    | 0    | -2             | -2     | -3   |
| Cyprus         |                  | 36   | -1   |                | 20     | -2   |
| Latvia         |                  | 0    | -1   |                | -3     | -3   |
| Lithuania      |                  | -16  | 0    |                | 0      | -2   |
| Luxembourg     | -14              | -11  | -12  | 2              | 2      | 0    |
| Hungary        |                  | 0    | -2   |                | -1     | -2   |
| Malta          |                  | 8    | -35  |                | 125    | -2   |
| Netherlands    | -4               | -3   | -3   | -1             | -2     | -2   |
| Austria        | -8               | -7   | -15  | -2             | -2     | 0    |
| Poland         |                  | 0    | -3   |                | 0      | -2   |
| Portugal       | -1               | 0    | 0    | -4             | 0      | -2   |
| Slovenia       |                  | -2   | -1   |                | 0      | -2   |
| Slovakia       |                  | 1    | 0    |                | 0      | -2   |
| Finland        | -7               | -8   | -7   | 0              | -1     | -3   |
| Sweden         | -1               | -3   | -4   | -3             | -6     | -6   |
| United Kingdom | -23              | -24  | -23  | -2             | -2     | -2   |
| European Union | -7.9             | -3.3 | -8.1 | 1.7            | -0.1** | -2.6 |

Source: QUIET database.

\* Not available data for the 10 new Member States before 2004

\*\* Without Malta-outlier for 2004

## 2 Accuracy

The aim of this paragraph is not necessarily to provide figures on the accuracy of international trade statistics, but more realistically to inform users about the problems faced and existing limitations in ensuring accurate information. Where measures of accuracy are known, they are included. This section also informs users of the work being undertaken to improve accuracy, either nationally in the MSs or at Eurostat level.

### 2.1 Accuracy characteristics

#### 2.1.1 Exclusions

There are certain items that are excluded from the official international trade statistics, e.g. items that are of no commercial value. Intra-EU trade statistics do not cover arrivals and dispatches affected by private individuals or small enterprises, which are

exempt from periodic tax declarations. A list of exclusions can be found in Commission Regulation (EC) N. 1982/2004 for Intrastat and N. 1917/2000 for Extrastat.

## 2.1.2 Levels of thresholds

### Intrastat

In order to reduce the burden on enterprises, particularly small and medium sized ones, the Intrastat system is designed in a way that the workload for providers of information varies according to the amount of trade in which they are engaged. To achieve this, each MS applies a system of thresholds aimed at guaranteeing a satisfactory quality of statistical data, in particular to meet the coverage criteria required by Intrastat regulations, while limiting the workload for traders. Companies below the threshold do not have to declare at all (exclusion threshold), or they are allowed to send in a simplified declaration (simplification threshold).

Each MS sets these thresholds independently, for both arrivals and dispatches, in October before the year of trade. Tables 2 and 3 show, respectively, the level of exclusion and simplification thresholds for the years 2003 to 2005.

**Table 2: Intrastat Exclusion thresholds in EUR (No declaration below the threshold), 2003-2005**

| MEMBER STATES  | Arrivals |         |         | Dispatches |         |         |
|----------------|----------|---------|---------|------------|---------|---------|
|                | 2003     | 2004    | 2005    | 2003*      | 2004    | 2005    |
| Belgium        | 250 000  | 250 000 | 250 000 | 250 000    | 250 000 | 250 000 |
| Czech Republic |          | 62 500  | 62 500  |            | 125 000 | 125 000 |
| Denmark        | 201 613  | 201 613 | 201 613 | 336 022    | 336 022 | 604 027 |
| Germany        | 200 000  | 200 000 | 300 000 | 200 000    | 200 000 | 300 000 |
| Estonia        |          | 63 890  | 63 890  |            | 63 890  | 63 890  |
| Greece         | 29 000   | 29 000  | 29 000  | 44 000     | 44 000  | 44 000  |
| Spain          | 100 000  | 120 000 | 130 000 | 100 000    | 120 000 | 130 000 |
| France         | 100 000  | 100 000 | 100 000 | 100 000    | 100 000 | 100 000 |
| Ireland        | 190 000  | 191 000 | 191 000 | 635 000    | 635 000 | 635 000 |
| Italy(*)       | 103 291  | 150 000 | 150 000 | 154 937    | 200 000 | 150 000 |
| Cyprus         |          | 26 921  | 27 680  |            | 26 921  | 27 680  |
| Latvia         |          | 28 316  | 56 915  |            | 19 374  | 78 258  |
| Lithuania      |          | 13 612  | 26 066  |            | 28 093  | 46 339  |
| Luxembourg     | 100 000  | 100 000 | 150 000 | 100 000    | 100 000 | 150 000 |
| Hungary        |          | 100 000 | 160 00  |            | 100 000 | 400 000 |
| Malta          |          | 9 320   | 700     |            | 9 320   | 700     |
| Netherlands    | 225 000  | 225 000 | 400 000 | 225 000    | 225 000 | 400 000 |
| Austria        | 200 000  | 250 000 | 250 000 | 200 000    | 250 000 | 250 000 |
| Poland         |          | 61 543  | 69 565  |            | 102 573 | 115 942 |
| Portugal       | 60 000   | 60 000  | 60 000  | 85 000     | 85 000  | 85 000  |
| Slovenia       |          | 100 000 | 100 000 |            | 100 000 | 100 000 |
| Slovakia       |          | 72 305  | 72 305  |            | 144 610 | 144 610 |
| Finland        | 100 000  | 100 000 | 100 000 | 100 000    | 100 000 | 100 000 |
| Sweden         | 164 727  | 164 727 | 243 293 | 164 727    | 164 727 | 497 644 |
| United Kingdom | 364 000  | 323 114 | 323 114 | 364 000    | 323 114 | 323 114 |

Source: QUIET database.

\* companies below the threshold have to make a yearly simplified Intrastat declaration

**Table 3: Intrastat Simplification thresholds in EUR (Simplified declaration below the threshold), 2003-2005**

| MEMBER STATES | Arrivals |         |         | Dispatches |         |         |
|---------------|----------|---------|---------|------------|---------|---------|
|               | 2003     | 2004    | 2005    | 2003       | 2004    | 2005    |
| France        | 230 000  | 230 000 | 230 000 | 460 000    | 460 000 | 460 000 |
| Luxembourg    | 375 000  | 375 000 | 375 000 | 375 000    | 375 000 | 375 000 |
| Slovakia      |          | 299 832 | 310 884 |            | 749 580 | 777 210 |

Source: QUIET database.

### **Extrastat**

Until 2001, according to Commission Regulation (EC) N. 1917/2000 MSs process declarations exceeding the statistical threshold of 800 Euro or 1.000 kg. Thresholds rose to 1.000 Euro or 1.000 kg starting from January 2002 (Commission Regulation N. 1669/2001).

### **2.1.3 Adjustment rate for trade below threshold and non response**

#### **Intrastat**

After the annual thresholds have been set, estimates should be made for the trade below threshold. Statistical practices are different among the MSs, but are most often based upon the use of the fiscal information. However, not all the MSs estimate the “below the threshold trade” and “non-response”. Actually, four MSs (Greece, Spain, France and Malta) send figures to Eurostat without including adjustments on below the thresholds transactions. Eight MSs (Greece, Spain, France, Italy, Malta, Portugal, Slovenia and Finland) send figures to Eurostat without including adjustments on non-response, for different reasons.

Table 4 shows the share of adjustment for trade below the thresholds, calculated in terms of trade value, separately for arrivals and dispatches; the methodology used for the adjustment, if any, is also briefly described. In some cases, figures for the adjustments have not been provided. As can be seen, in general the adjustments are between 1% and 4% but they are quite varying between the MSs and sometimes even in time, for the same MS.

**Table 4: Method of adjustments and share of trade value below the thresholds in intra-EU trade, 2002-2004**

| MEMBER STATES         | Method of adjustment                                                                | Share (%) of value below the threshold estimated (last known value) |       |        |            |       |        |
|-----------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------|--------|------------|-------|--------|
|                       |                                                                                     | Arrivals                                                            |       |        | Dispatches |       |        |
|                       |                                                                                     | 2002*                                                               | 2003* | 2004** | 2002*      | 2003* | 2004** |
| <b>Belgium</b>        | Adjustment based on VAT data and constant sample method                             | 1.9                                                                 | 1.9   | 1.6    | 0.8        | 0.80  | 0.60   |
| <b>Czech Republic</b> | Adjustment based on historical figures                                              |                                                                     |       | 1.05   |            |       | 1.06   |
| <b>Denmark</b>        | Adjustment based on VAT data                                                        | 2.5                                                                 | 2.7   | 2.6    | 1.70       | 1.70  | 1.60   |
| <b>Germany</b>        | Adjustment based on VAT data                                                        | 2.74                                                                | 2.74  | 2.74   | 1.15       | 1.57  | 1.57   |
| <b>Estonia</b>        | Adjustment based on historical figures and VAT data                                 |                                                                     |       | 3.2    |            |       | 1.10   |
| <b>Greece</b>         | No adjustment                                                                       | -                                                                   | -     | -      | -          | -     | -      |
| <b>Spain</b>          | No adjustment                                                                       | -                                                                   | -     | -      | -          | -     | -      |
| <b>France</b>         | No adjustment in the years under analysis                                           | -                                                                   | -     | -      | -          | -     | -      |
| <b>Ireland</b>        | Adjustment based on VAT data or Holt-Winter's method if VAT not available           | 2.5                                                                 | 5.5   | 4.5    | 1.10       | 2.10  | 1.70   |
| <b>Italy</b>          | Adjustment based on "just-above-the-threshold" traders                              | 2.8                                                                 | 3.3   | 3.4    | 2.2        | 2.7   | 2.7    |
| <b>Cyprus</b>         | Adjustment based on historical figures and VAT data                                 |                                                                     |       | 0.7    |            |       | 0.10   |
| <b>Latvia</b>         | Adjustment based on historical figures and VAT data                                 |                                                                     |       | 3.15   |            |       | 4.00   |
| <b>Lithuania</b>      | Adjustment based on VAT data                                                        |                                                                     |       | 0.5    |            |       | 0.50   |
| <b>Luxembourg</b>     | Adjustment based on historical figures and VAT (together with non-response adjust.) | na                                                                  | na    | na     | na         | na    | na     |
| <b>Hungary</b>        | Adjustment based on VAT data                                                        |                                                                     |       | 1.9    |            |       | 1.10   |
| <b>Malta</b>          | No adjustment                                                                       |                                                                     |       | -      |            |       | -      |
| <b>Netherlands</b>    | Adjustment based on historical figures and VAT data                                 | na                                                                  | na    | na     | na         | na    | na     |
| <b>Austria</b>        | Adjustment based on VAT/VIES data                                                   | 3.0                                                                 | 3.5   | 2.2    | 0.70       | 1.20  | 2.20   |
| <b>Poland</b>         | Adjustment based on historical figures                                              |                                                                     |       | 1.22   |            |       | 1.29   |
| <b>Portugal</b>       | Adjustment based on VAT data                                                        | 2.1                                                                 | 2.2   | 2.0    | 1.10       | 1.20  | 1.20   |
| <b>Slovenia</b>       | Adjustment based on VAT data                                                        |                                                                     |       | 3.49   |            |       | 2.69   |
| <b>Slovakia</b>       | Adjustment based on VAT data                                                        |                                                                     |       | 2.29   |            |       | 2.89   |
| <b>Finland</b>        | Adjustment based on VAT data                                                        | 3.8                                                                 | 3.4   | 3.4    | 1.00       | 1.10  | 1.40   |
| <b>Sweden</b>         | Adjustment based on VAT of "just-above-the-threshold" traders                       | 2.7                                                                 | 2.4   | 2.2    | 1.50       | 1.20  | 1.10   |
| <b>United Kingdom</b> | Adjustment based on VAT of "just-above-the-threshold" traders                       | 2.6                                                                 | 2.8   | 2.6    | 2.5        | 2.9   | 3.0    |

Source: QUIET database.

na: not available

\* : Not available for the 10 new Member States before 2004

\*\* : For the 15 old Member States, data are relating to January – December 2004. For the 10 new Member States, data are relating to May - December 2004

Like the preceding Table 4, Table 5 shows the share (in terms of trade value) and description of the methodology in the case of adjustments for non-response.

The phenomenon of late or non-response by traders is a serious weakness in Intrastat. The rate of non-response has decreased since 1993. The problem of non-response can for some MSs be quite worrying and, in most of the cases, figures do not show improvements in the years under analysis.

Finally, information in Table 5 deals with annual figures, but it may be pointed out that the problem of accuracy caused by "non-response" is higher at monthly level than

at annual level (e.g. “non-response“ at monthly level includes missing declarations that would be included later at annual level).

**Table 5: Method of adjustments and share of non-response in intra-EU trade, 2002-2004**

| MEMBER STATES         | Method of adjustment                                                                                                        | Share (%) of non-response estimated (last known value) |                   |                    |                   |                   |                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------|--------------------|-------------------|-------------------|--------------------|
|                       |                                                                                                                             | Arrivals                                               |                   |                    | Dispatches        |                   |                    |
|                       |                                                                                                                             | 2002 <sup>*</sup>                                      | 2003 <sup>*</sup> | 2004 <sup>**</sup> | 2002 <sup>*</sup> | 2003 <sup>*</sup> | 2004 <sup>**</sup> |
| <b>Belgium</b>        | Adjustment based on VAT data and constant sample method (growth estimators are calculated to determine the trend)           | 1.2                                                    | 1.2               | 1.3                | 1.1               | 1.1               | 1.1                |
| <b>Czech Republic</b> | Adjustment using imputation method using historical data and using collected data of current month for indices computations |                                                        |                   | 1.68               |                   |                   | 1.65               |
| <b>Denmark</b>        | Adjustment based on VAT data                                                                                                | 10.5                                                   | 10.3              | 9.5                | 6.6               | 5.2               | 4.5                |
| <b>Germany</b>        | Adjustment based on VAT data                                                                                                | 0.87                                                   | 2.32              | 2.01               | 0.25              | 1.09              | 0.79               |
| <b>Estonia</b>        | Forecast based on past data (historical company data) or adjustment based on VAT data                                       |                                                        |                   | 4.37               |                   |                   | 5.6                |
| <b>Greece</b>         | No adjustment (non-response is expected to be insignificant)                                                                | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>Spain</b>          | No adjustment                                                                                                               | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>France</b>         | No adjustment in the years under analysis                                                                                   | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>Ireland</b>        | Forecast based on past data or, failing this, adjustment based on VAT data                                                  | 1.0                                                    | 0.7               | 1.00               | 0.3               | 0.4               | 0.5                |
| <b>Italy</b>          | No adjustment (survey is total)                                                                                             | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>Cyprus</b>         | Forecast based on past data (historical company data)                                                                       |                                                        |                   | 0.5                |                   |                   | 9.56               |
| <b>Latvia</b>         | Forecast based on past data (historical company data) or adjustment based on VAT data                                       |                                                        |                   | 1.65               |                   |                   | 0.9                |
| <b>Lithuania</b>      | Adjustment based on VAT data                                                                                                |                                                        |                   | 0.2                |                   |                   | 0.1                |
| <b>Luxembourg</b>     | Overall adjustment for below threshold trade and non-response or forecast based on past data (historical company data)      | <i>na</i>                                              | <i>na</i>         | <i>na</i>          | <i>na</i>         | <i>na</i>         | <i>na</i>          |
| <b>Hungary</b>        | Forecast based on individual time series of traded values declared by each PSI                                              |                                                        |                   | 3.8                |                   |                   | 8.6                |
| <b>Malta</b>          | No adjustment                                                                                                               |                                                        |                   | -                  |                   |                   | -                  |
| <b>Netherlands</b>    | Imputation method using historical data and adjustment method using weighting                                               | 18.0                                                   | 17.0              | 15.00              | 13.0              | 12.0              | 10.0               |
| <b>Austria</b>        | Adjustment method using weighting & imputation method using historical data or adjustment based on VAT/VIES data            | 2.7                                                    | 2.7               | 2.70               | 1.3               | 1.3               | 1.3                |
| <b>Poland</b>         | Forecast based on past data (historical company data)                                                                       |                                                        |                   | 5.37               |                   |                   | 2.78               |
| <b>Portugal</b>       | Adjustment based on common respondents, but not transmitted to Eurostat.                                                    | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>Slovenia</b>       | No adjustment (non-response is negligible)                                                                                  |                                                        |                   | -                  |                   |                   | -                  |
| <b>Slovakia</b>       | Forecast based on past data (historical company data) or adjustment based on VAT data                                       |                                                        |                   | 2.26               |                   |                   | 1.36               |
| <b>Finland</b>        | The effect of non-response is insignificant                                                                                 | -                                                      | -                 | -                  | -                 | -                 | -                  |
| <b>Sweden</b>         | Adjustment based on previous data and on VAT data                                                                           | 1.3                                                    | 0.9               | 0.9                | 0.7               | 0.6               | 0.9                |
| <b>United Kingdom</b> | Constant sample method (growth estimators are calculated to the trend)                                                      | 0.11                                                   | 0.06              | <i>na</i>          | 0.22              | 0.15              | <i>na</i>          |

Source: QUIET database.

na: not available

-: no adjustment for non-response

\*: Not available for the 10 new Member States before 2004

\*\* : For the 15 old Member States, data are relating to January – December 2004. For the 10 new Member States, data are relating to May - December 2004

According to Table 5, the shares of adjustments are bigger on arrivals than on dispatches, with exception of the three new Member States, Cyprus, Estonia and Hungary. This asymmetrical impact combined with the existence of MSs not adjusting their figures may suggest that the arrivals at EU level are likely to be underestimated in comparison with the dispatches.

### Extrastat

The problem of non-response for extra-EU trade should theoretically not exist since extra-EU trade statistics are based on customs declarations.

#### 2.1.4 Estimation of statistical value

Trade figures are published as “statistical value”. This is the value of goods at the border of the declaring country: FOB (free on board) value for export/dispatch or CIF (cost, insurance and freight) value for import/arrival. It may differ from the amount agreed on the sales agreement (the invoice value) as a result of the delivery terms used in the transaction. MSs usually apply a threshold to exempt traders from declaring the statistical value.

As Table 6 shows, most of the MSs collect invoice value and estimate the statistical value in intra-EU trade while some MSs collect invoice value and do not carry out adjustments on it in order to estimate the statistical value. Therefore, traders’ calculations and MSs differing methodologies in the adjustment may cause differences in the published figures.

**Table 6: Methods adopted for estimating the statistical value in intra-EU trade**

| MEMBER STATES         | Method of adjustment                                                 |
|-----------------------|----------------------------------------------------------------------|
| <b>Belgium</b>        | No adjustment - use of invoice value below threshold                 |
| <b>Czech Republic</b> | Correcting coefficient computed from historical data                 |
| <b>Denmark</b>        | Correcting coefficient computed by special surveys                   |
| <b>Germany</b>        | Correcting coefficient computed from above the threshold data        |
| <b>Estonia</b>        | Correcting coefficient computed from above the threshold data        |
| <b>Greece</b>         | No adjustment - statistical value always collected                   |
| <b>Spain</b>          | No adjustment - use of invoice value below threshold                 |
| <b>France</b>         | No adjustment- use of invoice value below threshold                  |
| <b>Ireland</b>        | Correcting coefficient computed from historical data below threshold |
| <b>Italy</b>          | Coefficient, computed from above the threshold data                  |
| <b>Cyprus</b>         | Correcting coefficient computed from above the threshold data        |
| <b>Latvia</b>         | Correcting coefficient computed from historical data                 |
| <b>Lithuania</b>      | No adjustment - use of invoice value below threshold                 |
| <b>Luxembourg</b>     | Correcting coefficient computed from above the threshold data        |
| <b>Hungary</b>        | Correcting coefficient computed from above the threshold data        |
| <b>Malta</b>          | No adjustment                                                        |
| <b>Netherlands</b>    | Correcting coefficient computed by special surveys                   |
| <b>Austria</b>        | No adjustment - use of invoice value below threshold                 |
| <b>Poland</b>         | Correcting coefficient computed from above the threshold data        |
| <b>Portugal</b>       | Correcting coefficients computed from above the threshold data       |
| <b>Slovenia</b>       | No adjustment - collect statistical value                            |
| <b>Slovakia</b>       | Correcting coefficient computed by special surveys                   |
| <b>Finland</b>        | Correcting coefficient computed from above the threshold data        |
| <b>Sweden</b>         | Correcting coefficient computed by special surveys                   |
| <b>United Kingdom</b> | Correcting coefficient computed by special surveys                   |

Source: QUIET database.

## 2.1.5 Revisions

Trade figures can be updated several times between the first sending to Eurostat and the subsequent revisions and publications at national and EU level.

Studies carried out on the updating process of Intra and Extra-EU total trade data available in Eurostat shows that updates of the monthly statistics may be considerable and affect the quality of data. The monthly average, calculated on values, varies significantly according to MS and exceptionally may exceed 10% over a year.

The average extent of the revisions on the EU aggregates is generally higher for intra-EU trade than for extra-EU trade, and is also higher for arrivals than for dispatches.

The updating process of Intra and Extra-EU statistics thus induces some uncertainty, particularly concerning monthly figures and short-term analysis on the development of the international trade statistics.

Table 7 includes results, referring to trade in value. These results reflect the most recent figures related to the impact of the revisions, obtained directly from data transmitted to Eurostat by MSs. Signs are indicative of the direction of this impact. Negative signs state higher first figure whereas positive signs indicate smaller first figure than last one.

**Table 7: Impact of revisions (in %) in intra- and extra-EU trade, 2003-2004**

| MEMBER STATES  | % of change between the 1st and final results (last update available) |        |            |        |                |        |         |        |
|----------------|-----------------------------------------------------------------------|--------|------------|--------|----------------|--------|---------|--------|
|                | Intra-EU trade                                                        |        |            |        | Extra-EU trade |        |         |        |
|                | Arrivals                                                              |        | Dispatches |        | Imports        |        | Exports |        |
|                | 2003*                                                                 | 2004** | 2003*      | 2004** | 2003*          | 2004** | 2003*   | 2004** |
| Belgium        | 1.2                                                                   | -1.8   | 1.8        | -1.1   | 9.1            | 9.8    | 9.7     | 7.1    |
| Czech Republic |                                                                       | 2.3    |            | 1.9    |                | 9.0    |         | 7.3    |
| Denmark        | -4.0                                                                  | -2.9   | -4.9       | -1.7   | 1.3            | 3.2    | 0.3     | 4.1    |
| Germany        | 0.3                                                                   | -0.3   | 0.7        | -0.3   | 1.2            | 0.1    | 0.8     | 0.6    |
| Estonia        |                                                                       | -4.5   |            | 0.0    |                | -20.9  |         | -0.6   |
| Greece         | 1.1                                                                   | -0.3   | 5.3        | 0.0    | 1.1            | 1.4    | -2.1    | 3.1    |
| Spain          | 6.0                                                                   | 5.0    | 4.3        | 2.9    | 1.7            | 2.4    | 2.1     | 2.4    |
| France         | 7.5                                                                   | 3.9    | 5.2        | 2.5    | 1.1            | 0.9    | 0.7     | 0.7    |
| Ireland        | 1.8                                                                   | 3.5    | 0.8        | 1.0    | -1.3           | -0.3   | -0.3    | -0.2   |
| Italy          | 4.1                                                                   | 4.0    | 4.6        | 4.3    | -0.1           | 0.1    | 0.0     | -0.3   |
| Cyprus         |                                                                       | -4.0   |            | 5.0    |                | -1.4   |         | 8.5    |
| Latvia         |                                                                       | 3.1    |            | 2.1    |                | 0.0    |         | 0.0    |
| Lithuania      |                                                                       | 3.7    |            | 5.6    |                | 0.7    |         | -5.7   |
| Luxembourg     | 1.5                                                                   | 5.0    | 1.2        | 0.3    | -2.5           | -3.6   | 8.7     | 11.3   |
| Hungary        |                                                                       | -0.6   |            | 5.0    |                | 22.6   |         | 17.8   |
| Malta          |                                                                       | 0.2    |            | 0.3    |                | 0.0    |         | 0.0    |
| Netherlands    | 1.3                                                                   | -0.2   | 0.6        | 0.0    | 2.0            | 2.3    | 1.5     | 0.2    |
| Austria        | 6.8                                                                   | 9.1    | 2.8        | 6.5    | 3.1            | 1.9    | 13.5    | 8.0    |
| Poland         |                                                                       | 6.8    |            | 3.6    |                | 3.7    |         | -0.1   |
| Portugal       | 18.5                                                                  | 6.7    | 11.9       | 6.4    | 0.6            | 0.4    | 0.2     | 1.4    |
| Slovenia       |                                                                       | 5.9    |            | 7.7    |                | 1.2    |         | 1.1    |
| Slovakia       |                                                                       | 1.0    |            | 0.3    |                | 0.1    |         | 0.0    |
| Finland        | 7.2                                                                   | 7.6    | 1.8        | 1.5    | 2.3            | 3.2    | 1.3     | 1.5    |
| Sweden         | 1.7                                                                   | 1.4    | 1.1        | 0.4    | 1.9            | 2.2    | 2.2     | 1.3    |
| United Kingdom | 4.8                                                                   | 5.6    | 4.1        | 2.8    | 0.1            | 0.1    | 0.3     | 0.2    |
| EU             | 3.6                                                                   | 2.5    | 2.4        | 1.4    | 1.5            | 1.6    | 1.6     | 1.3    |

Source: detailed data transmitted to Eurostat.

-: First figure>Last figure

+: First figure<Last figure

\*: Not available data for the 10 new Member States in 2003.

\*\*: For the new Member States, 2004 data are relating to the cumulative period from May to December 2004.

From table 7 follows that, in most of the cases there is an underestimation of the final transmitted figure. Final data is generally available one year after the end of the year it refers to (8-9 months for some MSs).

## 2.2 Controls using “primary” information/sources

Controlling and correcting errors is resource intensive and MSs need to make judgements about when to make a correction. MSs use different criteria and employ different methodologies; this can be seen as discrepancies when comparing two MSs’ mirror figures.

At the level of the declarations, controls can be broadly classified as:

- Basic controls: this means checks on the *validity* of data and normally concerns all the variables collected. A validity error is when data is not correct therefore it may cause processing problems, it cannot be included in the statistical analysis and so it must be removed or corrected. Examples of this kind of errors are incorrect or missing codes, missing values in the declarations, character data in numeric field and vice versa, etc. All MSs use reference tables with an automatic process to control these errors.
- Complex controls: this means checks on the *accuracy* of data and assesses all possible errors associated with the data. The data is valid according to the previous criterion of validity but it is possibly incorrect therefore it can be processed but, if included, it will distort the statistical analysis. The declaration for example is inconsistent internally in the comparison of the different variables, or it is not consistent with what was submitted in the previous months. The methods used by the MSs consist in checking one variable against the other within the declaration or between declarations. The differences found among the MSs are in the kind of ratios chosen to be checked.

Similar controls can in addition be carried out by the MSs at a more aggregated level, for example at the detail of product requested in the Regulations, before sending the data to Eurostat.

At Eurostat level, consistency checks (detection of wrong codes, consistency of the different level of aggregation, etc...) are made before data is loaded into the Comext database of international trade and before publication. In addition, Eurostat has implemented a method for the detection of outliers, at detailed product level, based on the ratios between three published figures (net mass, value and quantity).

## 2.3 Controls using “secondary” information/sources

MSs normally check their intra-EU data at least with the data from VAT. Other sources could also be used, for example Balance of Payments, data from business registers, statistics made by a particular professional federation, etc. The purpose of these secondary sources is to provide supplementary information to the trade data as reported by the Intrastat system. This secondary information can be used in checking the credibility of the data or as a verification of traders’ declarations.

At Eurostat level, no control using secondary information is undertaken. Nevertheless, mirror exercises are regularly made and provide Eurostat with an estimate of the margin of error related to intra-EU flows. Since the implementation of Intrastat in 1993, dispatches are systematically higher than arrivals (see Tables 11 and 12), whereas, arrivals should theoretically be slightly higher (1.0% - 2%) than dispatches.

## 2.4 Confidentiality

The precise operational criteria determining which statistical data are considered confidential are fixed by each MS in the light of national legislation or practice. In general, in international trade statistics the principle of “passive confidentiality” is applied, by which data should be confidentialised in a MS only at the request of traders who feel that their interests would be harmed by the dissemination of their data. Passive confidentiality is therefore different from “active confidentiality”, where it is the MS who takes directly the initiative in acting on a potentially disclosive situation, for example by hiding cells in tables where less than 3 traders appear or one is dominant according to a certain rule.

Confidentiality may affect all type of flow, it can concern both value and quantity variables, one or all partner countries, etc. The considerable amount of detail in trade data means that the potential for the creation of confidential data at detailed level is extremely high. Passive confidentiality allows minimising the effect of confidentialisation on the quality of detailed data. Aggregating data before publication can safeguard information relating to a trader or trading activity. Whether partly or completely suppressed, full data for suppressed commodity codes is re-introduced in higher level aggregations if it is no longer possible to identify or deduce data for an individual protected commodity code. This allows complete data to be available to users but under broad commodity headings or at grand total level.

We distinguish the following three types of confidentiality:

- Partner Confidentiality: if a MS wishes to conceal the destination or the source or origin of a product, the code of the partner country is replaced by a ‘secret country code’, different for intra- and extra-EU trade.
- Product confidentiality: a MS may be requested on the other hand to allocate all or part of the trade to a confidential product code and in doing so, to “hide” the nature of the commodity involved. In fact, information about a product may be regarded as commercially sensitive by the trader either for the value, the quantity or more frequently the ratio between these two, since it would give an indication of the price of the product.
- Product and Partner Confidentiality: the two preceding types are applied at the same time; therefore both the partner and the product are hidden.

The levels of these three types of confidentiality can be assessed using data available in Eurostat, where special codes are used for hiding partner or product. The figures are presented in the following Tables 8 and 9 and are calculated for trade in value.

Table 8 shows the evolution in time, for the years 2002 to 2004, of the weight of confidential trade over total trade for each flow, at MS level. As can be noticed, the importance of confidentiality is very different depending on the kind of trade, on the flow and on the MS applying it.

**Table 8: Impact of the confidentiality on EU trade (share of total value), 2002-2004**

| MEMBER STATES  | Intra-EU trade (%) |       |        |            |       |        | Extra-EU trade (%) |       |        |         |       |        |
|----------------|--------------------|-------|--------|------------|-------|--------|--------------------|-------|--------|---------|-------|--------|
|                | Arrivals           |       |        | Dispatches |       |        | Imports            |       |        | Exports |       |        |
|                | 2002*              | 2003* | 2004** | 2002*      | 2003* | 2004** | 2002*              | 2003* | 2004** | 2002*   | 2003* | 2004** |
| Belgium        | 0.32               | 0.28  | 0.41   | 2.79       | 2.46  | 2.20   | 1.11               | 1.13  | 1.24   | 4.28    | 3.95  | 3.69   |
| Czech Republic |                    |       | 0.92   |            |       | 0.90   |                    |       | 7.89   |         |       | 1.42   |
| Denmark        | 1.56               | 1.09  | 0.76   | 8.79       | 9.05  | 8.39   | 1.49               | 1.35  | 0.84   | 12.86   | 12.98 | 12.84  |
| Germany        | 1.66               | 1.68  | 1.49   | 3.64       | 3.76  | 3.60   | 4.42               | 5.10  | 5.01   | 3.40    | 3.62  | 3.52   |
| Estonia        |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Greece         | 0.34               | 0.22  | 0.90   | 2.21       | 3.09  | 3.38   | 2.39               | 0.89  | 0.10   | 0.71    | 0.76  | 1.04   |
| Spain          | 0.22               | 0.20  | 0.22   | 0.21       | 0.39  | 0.57   | 0.41               | 0.39  | 0.33   | 1.20    | 1.35  | 1.63   |
| France         | 0.00               | 0.00  | 0.00   | 1.93       | 1.98  | 1.96   | 0.01               | 0.01  | 0.01   | 1.75    | 1.76  | 1.63   |
| Ireland        | 0.06               | 0.09  | 0.05   | 0.07       | 0.06  | 0.03   | 0.10               | 0.04  | 0.24   | 0.00    | 0.00  | 0.00   |
| Italy          | 0.96               | 1.12  | 1.14   | 0.77       | 0.81  | 0.76   | 6.26               | 6.66  | 6.71   | 0.68    | 0.72  | 0.76   |
| Cyprus         |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Latvia         |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Lithuania      |                    |       | 0.03   |            |       | -      |                    |       | 1.42   |         |       | -      |
| Luxembourg     | 2.82               | 2.72  | 2.77   | 1.21       | 1.16  | 0.95   | 0.63               | 0.52  | 0.57   | 2.98    | 2.48  | 3.67   |
| Hungary        |                    |       | 1.43   |            |       | 0.76   |                    |       | 9.49   |         |       | 1.67   |
| Malta          |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Netherlands    | 2.07               | 2.08  | 0.89   | 4.74       | 4.73  | 4.83   | 2.86               | 2.60  | 2.59   | 8.04    | 7.79  | 2.87   |
| Austria        | 0.92               | 1.11  | 1.11   | 7.14       | 5.49  | 5.34   | 3.84               | 4.61  | 7.94   | 8.14    | 6.33  | 6.24   |
| Poland         |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Portugal       | -                  | -     | -      | 0.07       | 0.05  | 0.07   | -                  | -     | -      | 0.68    | 0.75  | 0.87   |
| Slovenia       |                    |       | -      |            |       | -      |                    |       | -      |         |       | -      |
| Slovakia       |                    |       | 0.19   |            |       | 0.31   |                    |       | 22.64  |         |       | 4.68   |
| Finland        | 0.09               | 0.10  | 1.05   | 0.34       | 0.42  | 6.91   | 0.33               | 0.35  | 6.13   | 0.04    | 0.17  | 5.26   |
| Sweden         | 0.16               | 0.32  | 0.25   | 3.88       | 3.93  | 3.32   | 0.36               | 0.32  | 0.49   | 3.39    | 4.11  | 4.67   |
| United Kingdom | 1.67               | 1.46  | 1.72   | 4.10       | 4.15  | 4.28   | 9.25               | 5.98  | 4.81   | 8.51    | 6.27  | 6.39   |
| European Union | 0.92               | 0.91  | 0.86   | 2.96       | 3.14  | 2.98   | 3.82               | 3.40  | 3.42   | 3.76    | 3.62  | 3.31   |

Source: Comext database; data extracted on 30/01/2006.

\* : Not available for the 10 new Member States before 2004

\*\* : For the 15 old Member States, data are relating to January – December 2004. For the 10 new Member States, data are relating to May - Decemeber 2004

- : No confidentiality applied

0.00: Confidentiality is applied but with marginal impact

It should also be noticed that the shares have been calculated for confidentiality in value at total product level only; the picture would be different if we were considering a more disaggregated product level, for example the chapters.

Table 9 splits the share of confidential data seen in Table 8 into partner confidentiality, product confidentiality and simultaneously partner and product confidentiality. As can be noticed, for most of the MSs, product confidentiality seems to affect rather more the dispatches/exports than the arrivals/imports at least at EU level. Moreover, some MSs are systematically only using one kind of confidentiality. At EU level, it seems that for arrivals the higher weight is due to secret partner while for imports, dispatches/exports it is due to secret product.

**Table 9: Impact of different type of confidentiality on EU trade (share of total value), 2004**

| MEMBER STATES  | Intra-EU trade (%) |                |                  |                |                |                  | Extra-EU trade (%) |                |                  |                |                |                  |
|----------------|--------------------|----------------|------------------|----------------|----------------|------------------|--------------------|----------------|------------------|----------------|----------------|------------------|
|                | Arrivals           |                |                  | Dispatches     |                |                  | Imports            |                |                  | Exports        |                |                  |
|                | Secret partner     | Secret product | Partner and prod | Secret partner | Secret product | Partner and prod | Secret partner     | Secret product | Partner and prod | Secret partner | Secret product | Partner and prod |
| Belgium        | -                  | 0.41           | 0.00             | -              | 2.19           | 0.00             | -                  | 1.21           | 0.04             | -              | 3.61           | 0.08             |
| Czech Republic | -                  | 0.88           | 0.03             | -              | 0.88           | 0.02             | -                  | 7.84           | 0.04             | -              | 1.24           | 0.18             |
| Denmark        | -                  | 0.24           | 0.52             | 0.15           | 3.06           | 5.18             | -                  | 0.41           | 0.43             | 0.06           | 3.38           | 9.40             |
| Germany        | 1.48               | 0.00           | 0.01             | 0.95           | 1.64           | 1.01             | 4.99               | 0.00           | 0.01             | 1.14           | 1.76           | 0.62             |
| Estonia        | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Greece         | -                  | -              | 0.90             | -              | -              | 3.38             | -                  | -              | 0.10             | -              | -              | 1.04             |
| Spain          | -                  | 0.22           | -                | -              | 0.57           | -                | -                  | 0.33           | -                | -              | 1.63           | -                |
| France         | 0.00               | -              | -                | 0.00           | 1.92           | 0.04             | 0.01               | -              | -                | 0.00           | 1.62           | 0.01             |
| Ireland        | -                  | 0.03           | 0.02             | -              | 0.02           | 0.02             | -                  | 0.23           | 0.01             | -              | 0.00           | 0.00             |
| Italy          | -                  | -              | 1.14             | -              | -              | 0.76             | -                  | -              | 6.71             | -              | -              | 0.76             |
| Cyprus         | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Latvia         | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Lithuania      | 0.03               | -              | -                | -              | -              | -                | 1.42               | -              | -                | -              | -              | -                |
| Luxembourg     | -                  | -              | 2.77             | -              | -              | 0.95             | -                  | -              | 0.57             | -              | -              | 3.67             |
| Hungary        | -                  | 1.43           | -                | -              | 0.76           | -                | -                  | 9.49           | 0.00             | -              | 1.67           | -                |
| Malta          | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Netherlands    | 0.30               | 0.55           | 0.03             | 0.40           | 3.78           | 0.65             | 0.38               | 1.21           | 1.00             | 0.77           | 1.33           | 0.77             |
| Austria        | 0.94               | 0.18           | -                | 1.81           | 3.50           | 0.02             | 6.91               | 1.03           | -                | 2.03           | 4.13           | 0.08             |
| Poland         | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Portugal       | -                  | -              | -                | -              | -              | 0.07             | -                  | -              | -                | -              | -              | 0.87             |
| Slovenia       | -                  | -              | -                | -              | -              | -                | -                  | -              | -                | -              | -              | -                |
| Slovakia       | -                  | 0.19           | -                | -              | 0.31           | -                | -                  | 22.64          | 0.00             | -              | 4.68           | 0.00             |
| Finland        | 0.28               | -              | 0.77             | 0.37           | -              | 6.54             | 0.31               | -              | 5.83             | 0.20           | -              | 5.06             |
| Sweden         | -                  | -              | 0.25             | -              | -              | 3.32             | -                  | -              | 0.49             | -              | -              | 4.67             |
| United Kingdom | 0.17               | 1.55           | -                | 0.15           | 4.13           | -                | 0.10               | 4.71           | 0.00             | 0.20           | 6.19           | -                |
| European Union | 0.36               | 0.33           | 0.16             | 0.34           | 1.95           | 0.69             | 1.16               | 1.30           | 0.96             | 0.44           | 2.06           | 0.82             |

Source: Comext database; data extracted on 30/01/2006.

For the 15 old Member States, data are relating to January – December 2004.

For the 10 new Member States, data are relating to May - Decemeber 2004

-: This type of confidentiality is not applied

0.00: This type of confidentiality is applied but with marginal impact

## 2.5 Actions to improve accuracy

At Eurostat level, the main approaches in order to improve accuracy of international trade statistics are:

- Methodological studies on controls, corrections, adjustments, asymmetries, revisions and confidentiality.
- Balancing approaches - trying to build the best possible EU25 and Euro-zone trade statistics using mirror statistics.
- Working group on data quality - trying to develop and promote more harmonised statistical practices among the MSs and in the view of the enlargement.

### 3 Accessibility & clarity

#### 3.1 Accessibility

Statistical data has the most value when it is easily accessible by all users under equal conditions and is available in the formats that users desire.

The following Table 10 shows, for each MS, the different means (hardcopy or electronic) and frequency of publications on international trade statistics.

**Table 10: Dissemination of international trade statistics**

| MEMBER STATES  | Hardcopy     |          |           |                       | Electronic                         |          |        |                          |
|----------------|--------------|----------|-----------|-----------------------|------------------------------------|----------|--------|--------------------------|
|                | News release | Bulletin |           | Other (yearly e.t.c.) | Online country bulletin / database | Diskette | CD ROM | Other (fax, email e.t.c) |
|                |              | monthly  | quarterly |                       |                                    |          |        |                          |
| Belgium        | X            | x        | x         | x                     | x                                  | x        | x      | x                        |
| Denmark        | X            | x        | x         | x                     | x                                  | x        | x      |                          |
| Germany        | X            | x        |           | x                     | x                                  |          | x      | x                        |
| Greece         | X            | x        |           | x                     | x                                  | x        |        | x                        |
| Spain          |              |          |           | x                     | x                                  |          | x      |                          |
| France         | X            | x        | x         | x                     | x                                  |          | x      | X                        |
| Ireland        | X            | x        |           | x                     | x                                  | x        | x      | X                        |
| Italy          | X            |          |           | x                     | x                                  | x        | x      | X                        |
| Luxembourg     |              |          |           | x                     | x                                  |          |        | X                        |
| Netherlands    |              | x*       | x*        |                       | x                                  |          | x*     |                          |
| Austria        | X            | x        | x         | x                     | x                                  |          | x      | X                        |
| Portugal       | X            |          |           | x                     | x                                  | x        | x      | x                        |
| Finland        | X            | x        |           | x                     | x                                  |          | x      | x                        |
| Sweden         | X            | x        | x         |                       | x                                  |          |        |                          |
| United Kingdom |              | x        | x         | x                     | x                                  | x        | x      | x                        |
| Cyprus         | X            | x        |           | x                     | x                                  |          |        |                          |
| Czech Republic | X            | x        | x         |                       | x                                  | x        | x      |                          |
| Estonia        | X            | x        |           | x                     | x                                  |          |        |                          |
| Hungary        | X            |          |           | x                     | x                                  |          | x      |                          |
| Latvia         | X            | x        | x         | x                     | x                                  | x        | x      |                          |
| Lithuania      | X            |          | x         | x                     | x                                  |          | x      | x                        |
| Malta          | X            |          | x         |                       | x                                  |          | x      |                          |
| Poland         |              | x        | x         | X                     | x                                  |          | x      |                          |
| Slovakia       | X            | x        |           |                       | x                                  | x        | x      |                          |
| Slovenia       | X            | x        | x         |                       | x                                  |          |        |                          |

Netherlands: \* only for regular tailored requests, will be phased out in coming years when data is available on their website.

Source: QUIET database (Questionnaire 2005).

## 3.2 Clarity

Clarity of the publications depends on the assistance provided to users by the producers of the data in using and interpreting statistics, and in comments and analysis of results available. The published statistics should carry with them the appropriate metadata, so that the user can understand and interpret them correctly.

Users of Eurostat international trade statistics can refer to the following information in order to understand and interpret data:

- User guide  
[http://epp.eurostat.ec.europa.eu/portal/page?\\_pageid=1073,46587259&\\_dad=portal&\\_schema=PORTAL&p\\_product\\_code=KS-BM-05-001](http://epp.eurostat.ec.europa.eu/portal/page?_pageid=1073,46587259&_dad=portal&_schema=PORTAL&p_product_code=KS-BM-05-001));
- Help desk on Comext database in Eurostat;
- Documentation under the responsibility or co-responsibility of units G2, G3 on the different nomenclatures used within EU international trade statistics:
  1. Combined Nomenclature (CN) - annual joint publication with DG Taxud;
  2. Explanatory Notes to the CN - joint publication with DG Taxud;
  3. Update of CN-codes (transposition tables etc. - annual updated publication by C 4) also on CD-ROM;
  4. Intrastat CN (only EN/FR available on paper for internal needs) - on CD-ROM (11 languages) - used by MSs for national publications;
  5. Geonomenclature - (Eurostat Theme 6 external trade) - annual publication.

## 4 Comparability

### 4.1 Comparability over space

Comparability in international trade statistics may be affected by different definitions and concepts used either in the MSs or in their main partners in the data collection.

#### **Comparability between EU international trade statistics and those of its main partners (Extra-EU trade)**

There can be differences in the methods used in the EU and those adopted by its trading partners in the rest of the world in terms of trade coverage, partner country, the value of transactions, etc. Thus, the EU bases its international trade statistics on the 'special trade' system, while the USA, Japan and Canada, for example, apply the 'general trade' system. Other differences may be due to the time lag (since the same operation can be recorded under a different reference month because of transport times), to statistical confidentiality applied only on one of the partners' trade, to different practices in terms of revisions and to problems of currency conversion.

These methodological differences can give rise to considerable statistical discrepancies in the data. Eurostat regularly performs reconciliations of EU trade statistics and those of its main trading partners (e.g. USA, Canada and Japan), in order to measure and explain the discrepancies.

Nevertheless, it should be stressed that considerable efforts have been made at international level to improve the harmonisation of methods. For example, in terms of product classifications, more than 150 countries use the Harmonised Commodity Description and Coding System (HS). Also, the United Nations publishes a handbook of methodological recommendations for compiling international trade statistics.

## Comparability between Community concept and National concept

Community legislation serves as a basis for compiling the extra- and intra-Community trade statistics sent by the MSs and published by Eurostat. However, Community statistics, which cover the EU as a whole, and the statistics compiled/published by the MSs, which are concerned with the national dimension, **are not always directly comparable**. MSs may in fact use a national concept to collect and publish their data but they have subsequently to adjust it in order to arrive at the Community concept requested by Eurostat.

There can be methodological differences, which make precise comparison of these two sets of statistics impossible. The principal differences are as follows:

### **- Breakdown by partner country**

For arrivals, certain MSs record the country of origin as the partner country, whereas the MS of consignment appears in Community statistics relating to the same movements.

### **- Treatment of goods in transit**

Some MSs, particularly Belgium and the Netherlands, do not record in their National figures goods, which they consider to be 'in transit'. This involves, firstly, imports from non-member countries which are cleared in these MSs before being dispatched to other MSs and, secondly, goods from other MSs which are immediately re-exported to non-member countries. These flows are included in the Community statistics under intra- or extra-EU trade, as appropriate. This phenomenon is sometimes referred to the 'Rotterdam effect'.

### **- Other differences**

Other methodological differences can cause discrepancies between national and Community statistics (examples: classification at national level as 'general trade' rather than 'special trade', or not recording repairs on the grounds that they are services).

The problem of CIF/FOB evaluation should play a negligible role in consideration of the geographical context and the structure on intra-EU trade in the MSs.

## **Intra-EU trade statistics based on Community concept and available in Eurostat**

In theory, intra-EU trade statistics based upon Community concept should be fully comparable; therefore data should generally be less affected by asymmetries than extra-EU. Export from Member State A to MS B, as reported by A, should be equal to import into B from A, as reported by B. However, since the Intrastat system came into operation, bilateral comparisons have revealed major and persistent discrepancies in the various MSs intra-Community trade statistics. Therefore, comparisons dealing with intra-EU trade statistics have to be made cautiously and should take into account the existence of these discrepancies.

The main reasons for the discrepancies are known and are represented by the thresholds, non-response and their related adjustments; statistical confidentiality; time lags in the registration of the transactions; misclassification of goods; other methodological differences.

The following Tables 11 and 12 report the intra-EU asymmetries at global level and by MS, calculated using trade in value.

**Table 11: Intra-EU asymmetries\* in %, 2002-2004**

|                    | 2002* | 2003* | 2004*<br>(Jan- Apr) | 2004**<br>(May-Dec) |
|--------------------|-------|-------|---------------------|---------------------|
| <b>Asymmetries</b> | 5.4   | 5.1   | 6.1                 | 3.0                 |

Source: Comext database; data extracted on 19/05/2006

\* Asymmetry = (EU 15 Dispatches – EU 15 Arrivals) / EU 15 Arrivals.

\*\* Asymmetry = (EU 25 Dispatches – EU 25 Arrivals) / EU 25 Arrivals.

Intra-EU asymmetries have always represented around 5% of trade since 1993 but, as can be noticed in Table 11, in recent years they have started to increase and represent in 2002 more than 6% of intra-EU trade. The figure for 2003 and 2004 shows global improvement.

Table 12 breaks down the asymmetries at MS level, separately for arrivals and dispatches. It is evident that the impact is very different among the MSs, it depends on the flow and it is quite varying on time. Eurostat regularly produces the so-called Mirror Leaflet to analyse the asymmetries. Many MSs regularly carry out bilateral studies to find out at detailed product level where the problems are and hopefully resolve them. However, remedies are not easily found despite all the analyses done.

**Table 12: Intra-EU asymmetries of arrivals/dispatches and their mirror figures in %, 2002-2004**

| MEMBER STATES         | Asymmetries for arrivals* |      |                    |                   | Asymmetries for dispatches** |       |                    |                   |
|-----------------------|---------------------------|------|--------------------|-------------------|------------------------------|-------|--------------------|-------------------|
|                       | 2002                      | 2003 | 2004<br>(Jan- Apr) | 2004<br>(May-Dec) | 2002                         | 2003  | 2004<br>(Jan- Apr) | 2004<br>(May-Dec) |
| <b>Belgium</b>        | -6.4                      | -4.7 | -5.2               | -6.6              | 16.3                         | 14.8  | 13.8               | 11.9              |
| <b>Czech Republic</b> |                           |      |                    | -18.1             |                              |       |                    | 27.9              |
| <b>Denmark</b>        | -7.5                      | -4.0 | -3.0               | -2.0              | 19.5                         | 13.5  | 12.0               | 9.6               |
| <b>Germany</b>        | 9.8                       | 7.2  | 10.0               | 5.8               | 0.8                          | 1.9   | 2.2                | -0.9              |
| <b>Estonia</b>        |                           |      |                    | -9.3              |                              |       |                    | -9.6              |
| <b>Greece</b>         | 31.8                      | 11.5 | 18.5               | 9.8               | -21.0                        | -13.7 | -2.3               | -11.4             |
| <b>Spain</b>          | 6.9                       | 8.6  | 7.5                | 6.4               | 5.5                          | 5.3   | 6.3                | 2.4               |
| <b>France</b>         | 6.6                       | 3.4  | 5.2                | 3.6               | 7.5                          | 9.1   | 10.1               | 5.4               |
| <b>Ireland</b>        | 11.9                      | 13.8 | 14.9               | 17.6              | 10.8                         | -4.6  | -4.0               | -6.7              |
| <b>Italy</b>          | 2.2                       | 2.9  | 2.5                | 0.6               | 12.1                         | 8.8   | 13.3               | 6.2               |
| <b>Cyprus</b>         |                           |      |                    | 9.1               |                              |       |                    | -49.9             |
| <b>Latvia</b>         |                           |      |                    | -6.1              |                              |       |                    | -30.7             |
| <b>Lithuania</b>      |                           |      |                    | -16.8             |                              |       |                    | 13.3              |
| <b>Luxembourg</b>     | 6.5                       | 14.1 | 17.2               | 15.6              | 7.2                          | -5.4  | -4.7               | -11.1             |
| <b>Hungary</b>        |                           |      |                    | -4.7              |                              |       |                    | 13.3              |
| <b>Malta</b>          |                           |      |                    | 4.3               |                              |       |                    | -10.0             |
| <b>Netherlands</b>    | 6.6                       | 5.4  | 6.2                | 6.5               | 7.9                          | 8.7   | 7.3                | 4.1               |
| <b>Austria</b>        | -0.7                      | -2.8 | -4.9               | -4.5              | 15.3                         | 15.8  | 22.5               | 11.3              |
| <b>Poland</b>         |                           |      |                    | -11.6             |                              |       |                    | 16.8              |
| <b>Portugal</b>       | 14.3                      | 14.4 | 19.6               | 17.8              | -5.3                         | -7.7  | -8.8               | -10.0             |
| <b>Slovenia</b>       |                           |      |                    | -3.6              |                              |       |                    | 3.3               |
| <b>Slovakia</b>       |                           |      |                    | -10.9             |                              |       |                    | 5.0               |
| <b>Finland</b>        | 2.6                       | 3.9  | 2.5                | 4.7               | 4.4                          | -1.9  | -3.6               | -6.1              |
| <b>Sweden</b>         | 7.8                       | 5.6  | 8.1                | 5.4               | -4.6                         | -2.7  | -2.2               | -2.7              |
| <b>United Kingdom</b> | 3.3                       | 8.2  | 9.3                | 8.7               | 8.9                          | 3.1   | 5.4                | -0.8              |

Source: Comext database; data extracted on 19/05/2006

\*: (Mirror dispatches – Arrivals) / Arrivals

\*\*: (Dispatches – Mirror arrivals) / Mirror arrivals

## 4.2 Comparability over time

Comparability over time is another important aspect of quality. Changes due to definitions, coverage or methods and other changes will have an impact on the continuity of international trade series.

The impact of the different methodological and practical changes that have occurred in recent years (changes in product nomenclatures, implementation of Intrastat, future EU enlargement, etc...) cannot be precisely assessed. Nevertheless, the development of intra-EU asymmetries at the moment of the introduction of Intrastat indicated, at the aggregated level and most importantly at more disaggregated level, that the implementation of Intrastat introduced a major break in the trade statistics time series.

## 5 Coherence

Coherence is defined as how well sets of statistics can be used together. Apart from the international trade statistics, information on trade flows can be found in:

- National Accounts,
- Business Statistics,
- Balance of Payments,

sources which must relate to one another.

However, the compilation of the data and the production of the above mentioned statistics follow the recommendations (sources and methods) of different international organisations, i.e. Eurostat, International Monetary Fund (IMF), Organisation for Economic Co-operation and Development (OECD), United Nations (UN), World Trade Organisation (WTO), etc.

Table 13 tries to give an overall idea of the main differences between these sources.

**Table 13: Coherence between International Trade Statistics, National Accounts, Balance of Payments and Business Statistics at the EU level**

|                                           | <b>Eurostat FTS</b>                   | <b>Eurostat National Accounts</b>                                                                                                      | <b>Eurostat BoP</b>                                                                                                                                                                                                                                                                               | <b>Business Statistics</b>              | <b>Conclusion</b>                                                                                       |
|-------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Concepts &amp; Definitions</b>         | Community definition                  | Aggregation of national statistics based on ESA 95. However, account of the rest of the world is still based upon national definition. | Aggregation of national statistics based upon IMF 5 <sup>th</sup> manual. It implies some methodological discrepancies (FOB/FOB, difference of coverage). Moreover, BoP statistics in Member States are most of the time derived from Member States' FTS that are based upon national definition. | Statistics are based on activity sector | Even if theoretical differences between these statistics are omitted, consistency cannot be guaranteed. |
| <b>Classifications (nomenclature)</b>     | Harmonised System (HS)                | NACE                                                                                                                                   | IMF classification                                                                                                                                                                                                                                                                                | NACE                                    | Compatibility between the existing nomenclature                                                         |
| <b>Aggregation level</b>                  | Detailed geographical breakdown       | Intra-EU , Extra-EU, World                                                                                                             | Detailed geographical breakdown                                                                                                                                                                                                                                                                   | All partners                            |                                                                                                         |
| <b>Statistical unit/object/population</b> | Declaration of an international trade |                                                                                                                                        | International Transaction                                                                                                                                                                                                                                                                         | Enterprise, local unit                  |                                                                                                         |
| <b>Reference period</b>                   | Monthly                               | Annual                                                                                                                                 | Annual                                                                                                                                                                                                                                                                                            | Annual                                  |                                                                                                         |
| <b>Correction methods</b>                 | National corrections                  | National corrections                                                                                                                   | National corrections                                                                                                                                                                                                                                                                              | National corrections                    | No harmonised practices can guarantee the coherence                                                     |

Source: internal documents.

## 6 Conclusions and way forward

As we have shown in the previous paragraphs, data quality in international trade statistics is very heterogeneous and some indicators' levels might be improved. MSs presented significant differences when looking at the various quality indicators chosen for this exercise up to 2004. It is clear that an important effort is requested to MSs in order to respect the different Regulations in force and assure a high quality of the data they produce. In this respect, an improvement of the Regulations in terms of data quality has also come into force since 2005.

In the 2005 quality report on International Trade Statistics have been included the respective indicators from the 10 new Member States (i.e. Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia and Slovenia). It is very important to have follow-up and updates of the quality information for all 25 EU MSs in order to improve the data quality of international trade statistics in the EU.

It is noteworthy that in most of the cases, the 10 new Member States have managed to come close to the average European Union level. Not great dissimilarities have been noticed between 10 new Member States and the former average European Union level.